PowerNap: Carbon-Aware Grid Intensity Forecasting

Tanvi Patel, Alexandra Lugo, Alex Muzila, Benny Yuan, Srivar Janna

Machine Learning for Climate Change
Professor Grace Lindsay

May 12, 2026

I. Abstract

As data centers become increasingly central to global computing infrastructure, their energy consumption and associated carbon emissions represent a growing environmental challenge. This paper presents PowerNap, a carbon intensity forecasting system designed to support carbon-aware compute scheduling across four major US electricity grid regions: California (CAISO), Texas (ERCOT), Illinois (MISO), and Virginia (PJM). Using hourly carbon intensity data from ElectricityMaps and weather features from Open-Meteo, we construct a unified dataset of 35,040 observations spanning the full year 2023 and train three model classes: Ridge regression, Random Forest, and a Long Short-Term Memory (LSTM) network implemented in PyTorch. We evaluate each model across two strategies: a chronological holdout split within each region and a leave-one-region-out generalization test designed to assess cross-region transferability. Our results show that while Random Forest and LSTM achieve lower error within regions seen during training, Ridge regression generalizes significantly better to held-out regions, suggesting that simpler linear models learn more transferable representations of grid dynamics. These findings have direct implications for the deployment of carbon-aware scheduling systems, where forecasts must be reliable across diverse and previously unseen grid conditions.

II. Background

The rapid growth of data center infrastructure, driven largely by the proliferation of artificial intelligence workloads, has made the energy consumption of computing systems a pressing climate concern. Data centers alone account for approximately 4% of global electricity consumption, and AI workloads such as training and serving large language models further amplify this impact (Guidi et al.). Since the carbon footprint of electricity consumption depends heavily on the instantaneous energy mix of the grid, the same compute workload can produce significantly different emissions depending on when and where it runs. This temporal and regional variability in grid carbon intensity creates an opportunity for emissions reduction through strategic scheduling.

The concept of carbon-aware computing has been explored most prominently by large technology companies with the infrastructure to act on it. Google's Carbon-Aware Computing system, described by Radovanovic et al. (2021), uses day-ahead carbon intensity forecasts combined with risk-aware optimization to generate hourly capacity limits for temporally flexible workloads across its global data center fleet, effectively delaying execution of batch jobs to greener time windows while preserving overall daily capacity. This work established the core principle underlying our project: that accurate short-term forecasting of grid carbon intensity is the technical prerequisite for any practical carbon-aware scheduling system

The problem of forecasting grid carbon intensity has been approached through a range of methods in the literature. Early work framed the problem as a time series regression task, noting that carbon intensity fluctuates based on power market mechanisms and weather conditions, and proposed statistical methods including LASSO and ARIMA as forecasting baselines (Leerbeck et al.). More recent work has shifted toward machine learning and deep learning approaches. Research has applied Long Short-Term Memory networks, hybrid models, and other deep learning architectures to capture temporal dynamics in carbon intensity series, though these methods tend to exhibit reduced accuracy in volatile grid conditions, particularly in regions with high renewable energy penetration (Tian et al.). The CarbonCast system demonstrated that hierarchical neural networks could forecast carbon intensity up to 96 hours in advance across 13 grid regions, achieving mean absolute percentage errors as low as 9.8% (Maji et al., "CarbonCast"). More recently, ensemble learning approaches have been applied to carbon intensity forecasting across multiple US grid regions including PJM and MISO, with results showing substantial inter-regional variability in prediction accuracy (Yan et al.).

Despite this body of work, most existing studies focus on forecasting accuracy within regions seen during training and do not rigorously evaluate how well models generalize to unseen grid regions. This is an important practical question: a carbon-aware scheduling system deployed by a company operating data centers across multiple regions must produce reliable forecasts for grids it may not have historical data from. Our work addresses this gap directly. Using publicly available data from ElectricityMaps and Open-Meteo, we train and compare three model classes, Ridge regression, Random Forest, and an LSTM, across four US grid regions with distinct energy mixes, and evaluate each model under both chronological holdout and leave-one-region-out generalization conditions. By doing so, we aim to assess not only which model forecasts most accurately within familiar regions, but which model class learns the most transferable representations of grid dynamics, a question with direct implications for real-world deployment of carbon-aware computing systems.

III. Methods

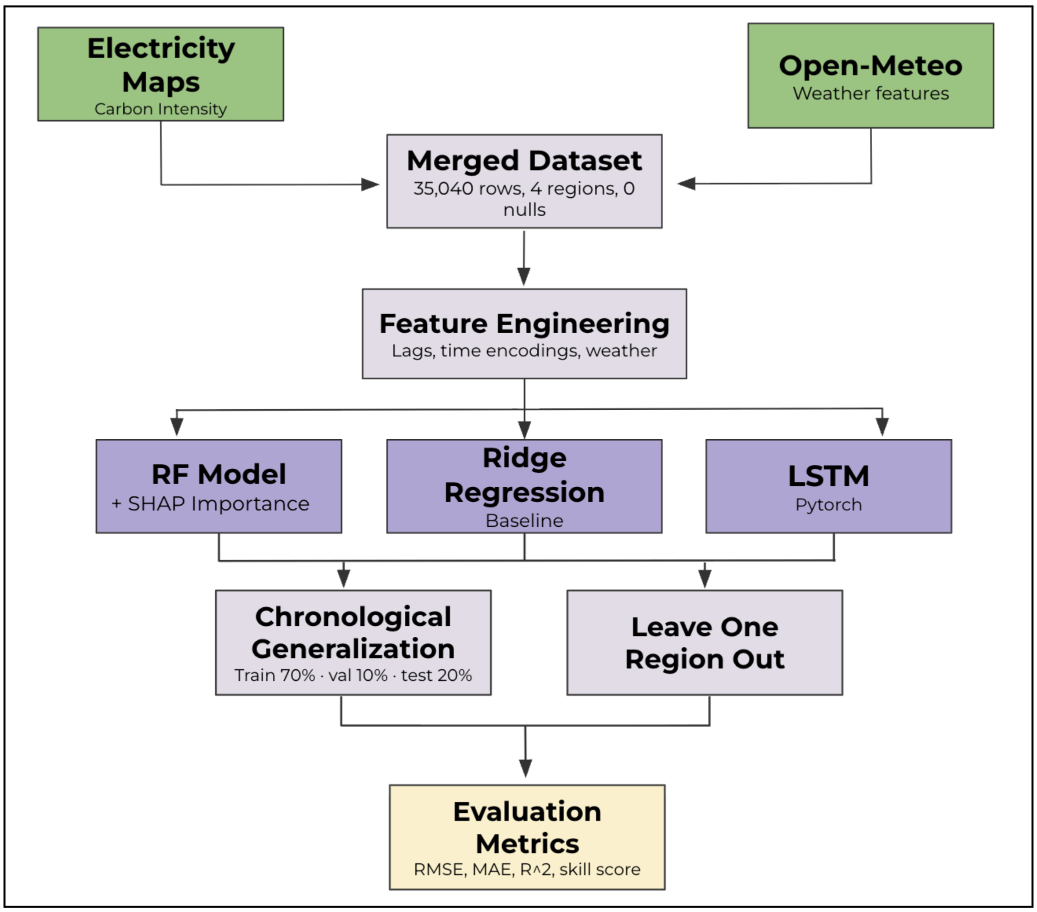

Fig 1. Overview of the PowerNap modeling pipeline.

3.1 Data Collection and Preprocessing

Carbon intensity data was sourced from ElectricityMaps at hourly resolution across four US grid regions: California (CAISO), Texas (ERCOT), Illinois (MISO), and Virginia (PJM). For each region, the dataset included direct carbon intensity in gCO2eq/kWh alongside two additional grid mix signals: carbon-free energy percentage (CFE%) and renewable energy percentage (RE%).

Weather data was pulled from Open-Meteo for the same four regions at matching hourly intervals, capturing four variables: temperature (°C), wind speed (km/h), cloud cover (%), and shortwave solar radiation (W/m²). The two sources were merged on datetime using an inner join, and the final dataset contained 35,040 rows with no missing values, covering the full calendar year 2023.

3.2 Feature Engineering

Four categories of features were constructed from the merged dataset. First, lag features were computed per region to give models access to recent historical carbon intensity: lags at 1, 2, 3, and 24 hours. The 24-hour lag specifically captures the same-hour-yesterday value, which is informative given the strong daily periodicity visible in the data. Second, cyclical time encodings were applied to hour of day, day of week, and month using sine and cosine transformations, so the model treats temporal boundaries (e.g., midnight-to-midnight, December-to-January) as continuous rather than discontinuous. Third, the four Open-Meteo weather variables were included directly. Fourth, the CFE% and RE% columns from ElectricityMaps provided a direct signal of grid composition at each timestep. After lag computation, rows containing NaN values (the initial window of each region's time series) were dropped. The final feature set contained 16 columns and was saved to a featured dataset used consistently across all three model families.

3.3 Ridge Regression

Ridge regression served as the interpretable baseline. A scikit-learn pipeline was constructed with standard scaling followed by a Ridge estimator using default regularization. For the chronological evaluation, a separate model was trained per region on that region's training split (trained on the first 70% of the year, validated on 10%, tested on the remaining 20%). For the leave-one-region-out evaluation, a single model was trained on the combined entries data of three regions and tested on the held-out region's full time series.

3.4 Random Forest

Random Forest was chosen to capture non-linear relationships and regional interactions that Ridge cannot model. The implementation used 50 estimators with a maximum depth of 8 across all models, parameters chosen to cap region-specific overfitting. Especially as true non-linear relationships exist in the real world (wind turbine cut-in speeds, and solar panel inverter cut-in thresholds), we expected an RF model to perform better than our baseline overall. The same two evaluation protocols were applied as for Ridge: per-region chronological models and cross-region leave-one-out models. Additionally, SHAP feature importance was computed post-hoc using one of the leave-one-out regional models (Virginia at 24 hours) as a diagnostic, though this was not part of the official cross-model comparison.

3.5 LSTM

The LSTM architecture was chosen because LSTMs are well suited for time-series forecasting tasks where current values depend on previous observations and recurring daily patterns. Two variants were trained: a regional model in which a separate LSTM was fit per region on that region's own chronological training split and tested on that region's held-out test period, and a global model that was trained on the combined data from three regions and tested on the remaining held-out region (leave-one-region-out). The architecture in both cases was a single-layer LSTM with 64 hidden units and a dropout rate of 0.2 applied to the output before a fully connected linear layer. The input to the model was a sequence of 24 consecutive hourly timesteps, each described by the 16 engineered features, resulting in input tensors of shape (batch, 24, 16). The model performed multi-task prediction, producing carbon intensity estimates simultaneously at the 1-hour, 6-hour, and 24-hour forecast horizons from a single forward pass. Features and targets were scaled independently using MinMaxScaler fit only on training data to prevent leakage. Training used the Adam optimizer at a learning rate of 1e-3 with MSE loss, a batch size of 32, and a maximum of 150 epochs. Early stopping with a patience of 20 epochs was applied using the validation loss, and the checkpoint with the lowest validation loss was restored at the end of training. Hyperparameters were initially evaluated through Weights and Biases sweeps, with the configuration above found to perform best.

3.6 Evaluation Strategy

Two evaluation protocols were applied consistently across all three model families. The first, chronological generalization, split each region's time series into train (70%), validation (10%), and test (20%) segments in strict temporal order, with a horizon-length boundary gap between train and validation to prevent label leakage. This tests whether a model trained on earlier data in a region can forecast later periods in that same region. The second protocol, leave-one-region-out, trained on three regions and evaluated on the fourth, cycling through all four regions as the holdout. This tests cross-regional generalization, specifically whether patterns learned from seen grids transfer to an unseen grid with a different energy mix.

3.7 Evaluation Metrics

Model performance was assessed using four metrics computed at each combination of model, region, and forecast horizon: mean absolute error (MAE), root mean squared error (RMSE), coefficient of determination (R^2), and a skill score defined relative to a naive persistence baseline. Additionally, a binary high/low carbon classification accuracy was computed by thresholding predictions against the regional median, providing a practical measure of whether the model correctly identifies clean versus carbon-heavy windows, which is the operationally relevant output for workload scheduling.

IV. Results

4.1 Evaluation Overview

The models' outputs across the prediction horizons are evaluated under two train/test split configurations: chronological and regional. The performance of each model under different train/test split configurations is measured by using metrics like mean absolute error (MAE) and R^2. We also assessed operational usefulness by testing whether models could identify periods of high and low carbon intensity based on the 75th and 25th percentile thresholds. The metrics produced by this test, such as precision and recall, are widely known and help non-technical stakeholders to understand the performance and reliability of the model in comparison to (MAE) and R^2.

4.2 Chronological Splits Performance

Chronological splits measure the models' ability to generalize across time. The MAE and R^2 shows a roughly inverse relationship. Lower MAE means more variance explained, which results in a natural high R^2 score. All three models show a similar trend in their prediction accuracy, which regresses as they predict into the future. The LSTM model shows slight dominance in performance in the 1-hour and the 6-hour prediction horizons, and the ridge model has a slight edge in the 24-hour prediction horizon.

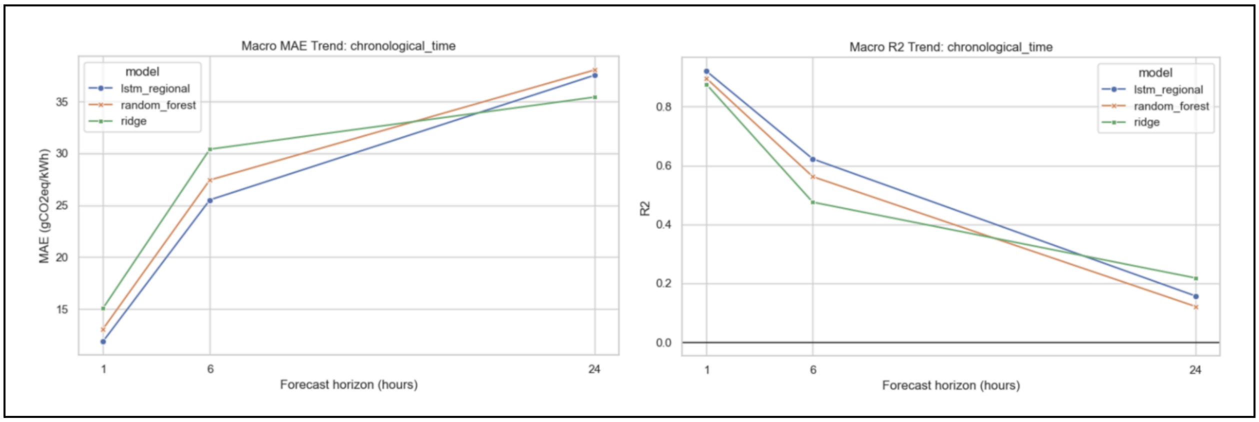

Fig. 2 shows both the macro MAE trend and the macro R2 trend for models trained and tested on a chronological time split. Macro MAE is calculated by averaging the mean absolute error across four different regions in the same horizon label. Macro R2 is calculated by averaging the R2 across four different regions in the same time horizon label.

To dig into the specific, Fig. 3 shows the models' performance by region and horizon. All three models have a fairly accurate presentation of Virginia's carbon intensity, while struggling to predict Texas's grid carbon intensity, especially for the 24-hour horizon.

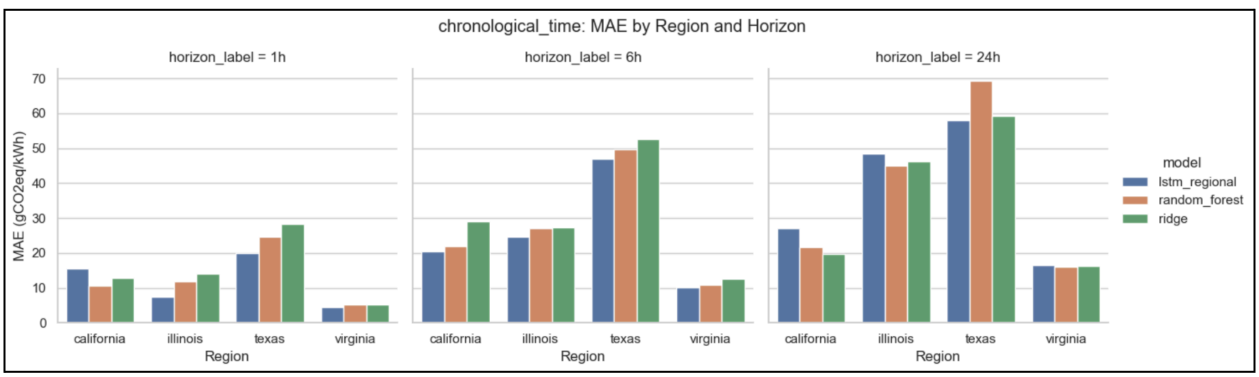

Fig. 3 shows the MAE by region and horizon. MAE is calculated by Avg(|y_pred-y_true|).

4.3 Regional Splits Performance

Regional splits are meant to test models' ability to generalize across regions. This is quite challenging for the models to do because, from EDA, we know that each region has a very different grid carbon emissions pattern. This expectation is reflected in the result. Fig. 4 shows that the random forest model is not very effective at generalizing across regions, and a simpler model, such as the ridge regression model, performs better in this kind of circumstance. If we look at the y-axis on the Macro MAE Trend graph for the regional split, we can see that the range is almost double compared to the same graph for the chronological split. Except for the ridge and the LSTM model on the 1-hour prediction horizon, all of the other model has a negative macro R^2 trend, meaning for each region the models fit the data worse than a horizontal line representing the mean of carbon intensity. The macro trend of MAE and R^2 emphasize the model's inability to generalize to out-of-distribution data. One interesting observation is the R^2 curve of the LSTM model, which achieved the best R^2 in the 1-hour prediction horizon but is the lowest for the other two prediction horizons.

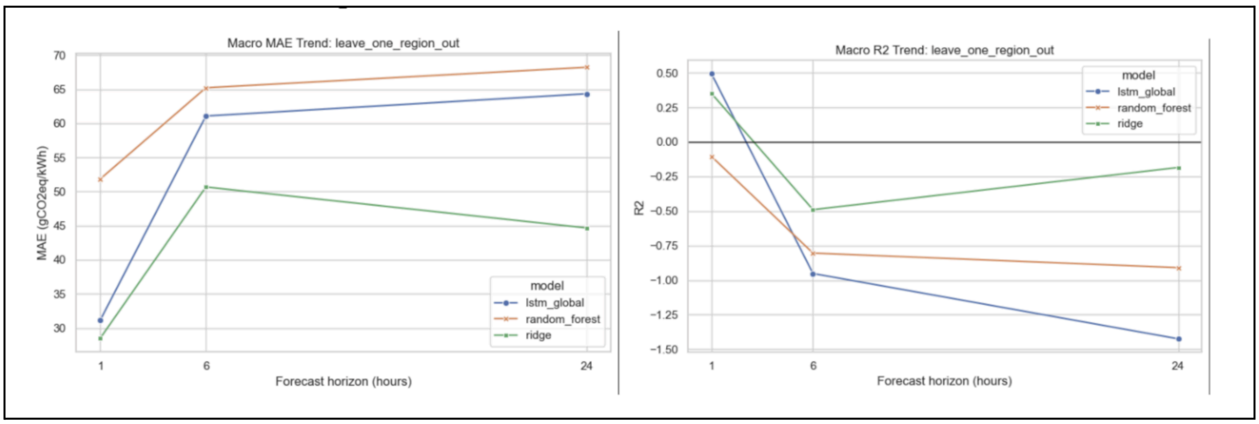

Fig. 4 shows both the macro MAE trend and the macro R2 trend for models trained and tested on a regional split. Metrics are calculated as the explanation from Fig. 2.

The detailed MAE graph by region and horizon also supported the macro trend. The LSTM and the random forest models have the hardest time generalizing to California's grid carbon intensity, indicating the distribution difference between the training set (Texas, Illinois, Virginia) and the testing set (California).

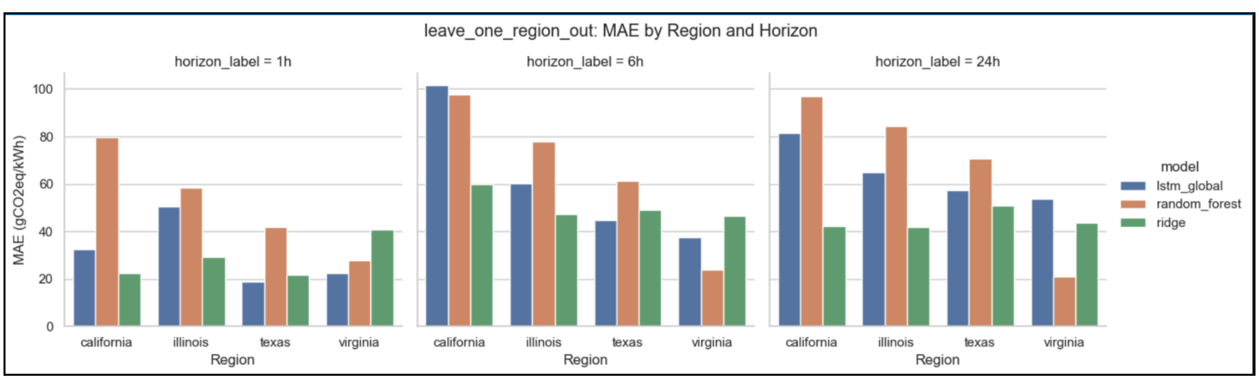

Fig. 5 MAE by region and horizon under the leave-one-region-out evaluation. Each model is trained on three regions and tested on the fourth. The LSTM and Random Forest show the largest errors on California, reflecting the distribution gap between California's solar-driven grid and the fossil-heavy training regions (Texas, Illinois, Virginia).

4.4 Comparison Between Temporal and Regional Evaluation Settings

Model	Horizon	Chronological MAE	Regional Holdout MAE	Difference
LSTM	1h	11.88	31.13	+19.26
Random Forest	1h	13.08	51.84	+38.76
Ridge	1h	15.07	28.50	+13.43
LSTM	24h	37.52	64.34	+26.82
Random Forest	24h	38.02	68.22	+30.19
Ridge	24h	35.41	44.67	+9.26
LSTM	6h	25.52	61.07	+35.56
Random Forest	6h	27.43	65.22	+37.78
Ridge	6h	30.40	50.68	+20.28

Fig. 6 Generalization gap by model and horizon. Chronological MAE is the macro-average error when testing on held-out time periods within the same region; Regional Holdout MAE is the macro-average error when testing on a held-out region. The difference column shows how much each model's performance degrades when the test region is unseen. Ridge has the smallest gaps across all three horizons.

Across models, regional holdout performance was generally weaker than chronological holdout performance, indicating that transferring across regions was more difficult than predicting later observations within the same year. This suggests that standard time-respecting evaluation may overestimate performance in applications where the model is deployed to new grid regions.

4.5 High- and Low-Carbon Period Prediction

To assess whether the models are useful for operational decision making, we evaluated their ability to identify high and low carbon periods rather than only their ability to minimize average prediction error. High carbon events were defined as hours above a region’s 75th percentile carbon intensity, while low carbon events were defined as hours below the 25th percentile. This event based evaluation makes the results more interpretable for carbon aware applications because it directly measures whether a model can support decisions such as avoiding carbon intensive periods or selecting lower carbon charging windows. Unlike MAE and R^2, which summarize continuous prediction error but may be difficult to interpret in practical terms, classification metrics such as accuracy, precision, and recall describe how reliably the model identifies the specific grid conditions that matter for deployment.

Fig. 7 shows that event classification performance was strongest under the chronological split, where all models achieved their highest accuracy at the 1-hour prediction horizon. Accuracy declined as the prediction horizon increased to 6 and 24 hours.

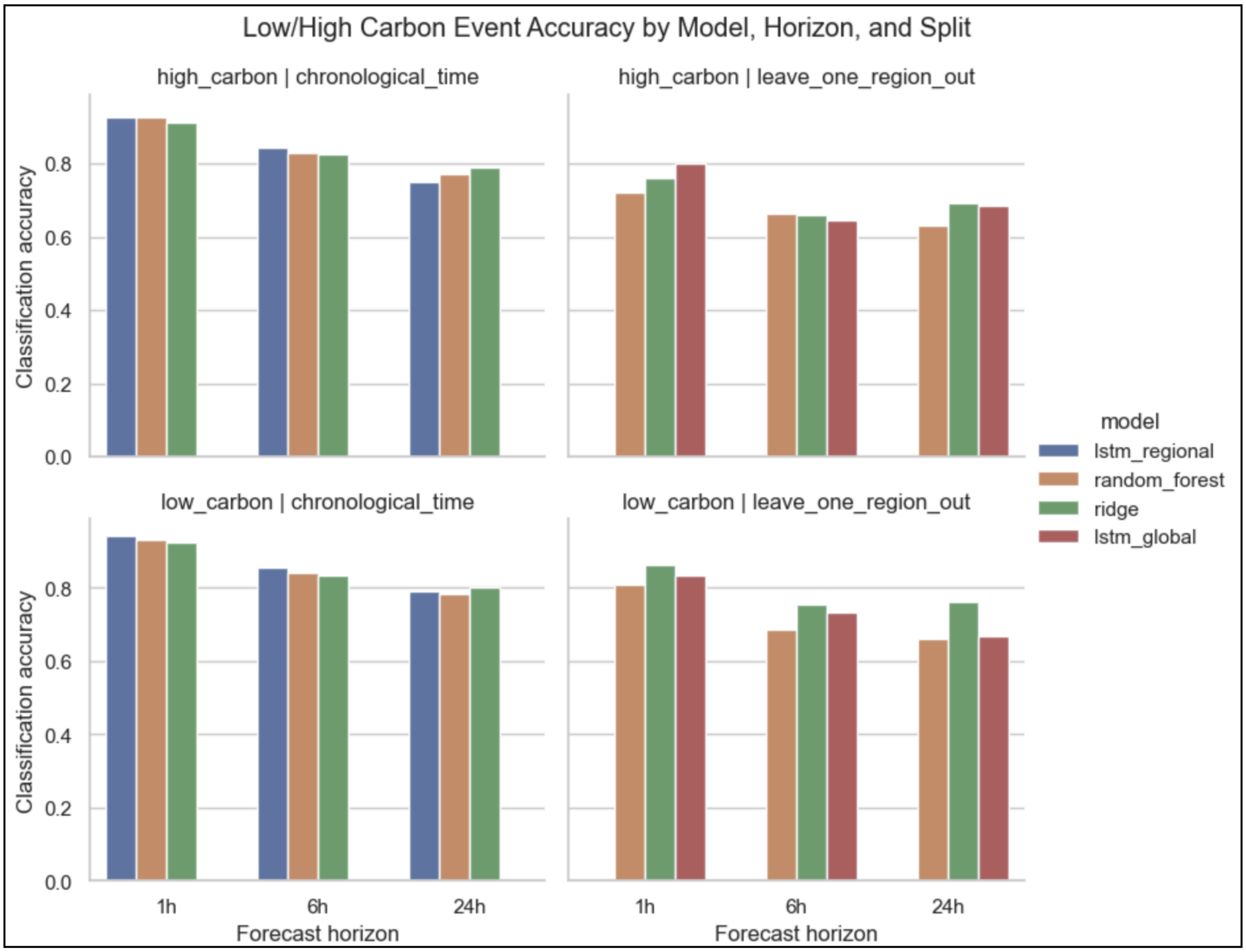

Fig. 7 High- and low-carbon classification accuracy by model, horizon, and split type. High-carbon periods are defined as hours above each region's 75th percentile carbon intensity; low-carbon periods are hours below the 25th percentile. Accuracy is highest at the 1-hour horizon under the chronological split and declines as the forecast horizon increases. Classification accuracy is more stable than MAE under the regional holdout setting, indicating that models retain practical scheduling utility even when point error is high.

Fig. 8 adds to the accuracy results by showing what kinds of errors the models make when identifying high and low carbon periods. These metrics matter because the cost of each mistake is different. For a high carbon period, a high FNR means the model fails to warn stakeholders about carbon-intensive hours, and a high FPR means the model recommends "clean" hours that are not actually low carbon, which violates our intention for this model: minimizing carbon emission through scheduling. Overall, error rates tend to increase as the prediction horizon gets longer. These metrics make it easier to choose a model based on the practical goal, such as avoiding high-carbon periods or selecting truly low-carbon windows.

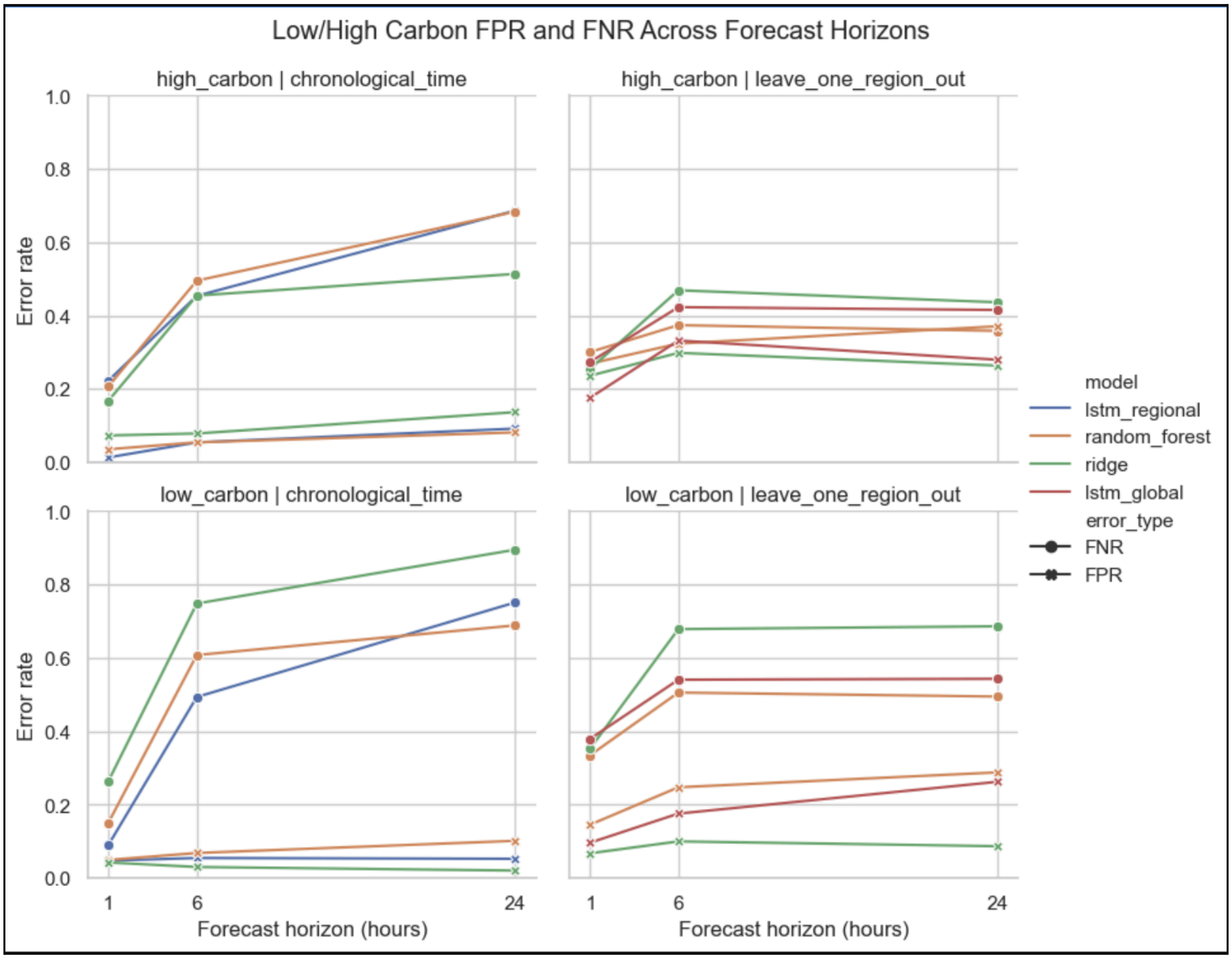

Fig. 8 False negative rate (FNR) and false positive rate (FPR) for high- and low-carbon period classification across forecast horizons and split types. A high FNR on high-carbon periods means the model fails to flag carbon-intensive hours; a high FPR on low-carbon periods means the model incorrectly labels dirty windows as clean. Both error types increase with horizon length. Ridge and the LSTM global model maintain the lowest error rates under the regional holdout setting.

V. Discussion

Within regions, all three models performed well at short horizons. The LSTM had the lowest 1-hour MAE (11.88 gCO2eq/kWh, R^2=0.966) and held that advantage at 6 hours (25.52). At 24 hours, Ridge had lower MAE (35.41) and higher R^2 (0.722) than both the LSTM (37.52, R^2=0.701) and Random Forest (38.02, R^2=0.647). Sequential modeling bought nothing at the long horizon, probably because the 24-hour lag already captures the most useful signal: same-hour-yesterday.

Cross-region generalization is where things diverge. Ridge held up: 1-hour MAE of 28.50 (R^2=0.889) on held-out regions, a generalization gap of 13.43 gCO2eq/kWh. Random Forest did not. Its 1-hour MAE jumped from 13.08 within-region to 51.84 cross-region, a gap of 38.76, and R^2 fell to 0.296 at 24 hours. The LSTM landed between them, with a 1-hour gap of 19.25 and R^2 bottoming at 0.402 at 6 hours. The pattern is the more a model memorizes regional structure, the harder it fails when the region changes. Ridge learns a smooth global weighting that transfers. Random Forest burns grid-specific non-linearities into its splits and cannot unlearn them.

Binary classification accuracy held up better than MAE across both settings. Since the actual goal is scheduling, defer or run, correctly identifying clean versus dirty windows matters more than point error. Most models got that right most of the time, even on unseen regions.

For a first-generation carbon-aware scheduler, these results suggest Ridge is the right starting point. It generalizes better than RF, requires no sequence construction, and trains in seconds. LSTM complexity is only justified if the deployment region appears in training data.

Three things would most improve this work. First, expanding beyond four US regions: all four share regulatory structure and data infrastructure, which makes generalization look easier than it is. Grids in Europe or Asia, with higher nuclear or coal shares, would challenge the models differently and likely require domain adaptation. Second, adding real-time grid mix features (nuclear percentage, hydro percentage) which are available from ElectricityMaps but were not included here. These are more structurally informative than weather proxies and would probably help cross-region transfer most. Third, training on multiple years. One year means one seasonal cycle, so year-specific anomalies cannot be separated from stable patterns.

The modeling gap worth exploring is Transformer architectures like Informer or PatchTST, which handle long-range dependencies more efficiently than LSTMs and have performed well on multi-horizon forecasting tasks. But the more pressing gap is deployment. An inference pipeline that queries live data and exposes scheduling recommendations via an API would turn this into something that actually runs jobs during clean grid windows, and it would also expose problems around data latency and real-time feature availability that offline evaluation cannot.

VI. Author Contributions & Disclosures

Name	**Contribution**
Tanvi Patel	Handled data engineering: pulling and cleaning hourly carbon intensity exports from ElectricityMaps for all four regions, collecting matching weather data from Open-Meteo, and writing the merging pipeline that produced the dataset used across all three models.
Alexandra Lugo	Built the feature engineering layer. This included the lag features, cyclical time encodings, and the train/test split logic shared by all tabular models.
Alex Muzila	Trained the Ridge regression baseline, designed the Random Forest models and ran SHAP analysis to examine which features most heavily impacted predictions. Created model comparison tables and preliminary visualizations to compare performance across different horizons.
Srivar Janna	Built the LSTM in PyTorch, covering sequence construction, MinMaxScaler normalization, multi-horizon output, and hyperparameter search via Weights and Biases. He trained both the regional and global variants.
Benny Yuan	Ran the final cross-model evaluation, produced the results visualizations and skill score comparisons, and

	coordinated the written report. He also built the carbon-aware scheduling simulation.

AI Use: Claude (Anthropic) was used to review and check work during the project. All code was written by the authors; no code was adapted from external sources.

References

ElectricityMaps Dataset Platform. ElectricityMaps, app.electricitymaps.com/datasets. Accessed May 2026.

Guidi, Gianluca, et al. "Environmental Burden of United States Data Centers in the Artificial Intelligence Era." *arXiv*, 14 Nov. 2024, arxiv.org/html/2411.09786v1.

Leerbeck, Kenneth, et al. "Short-Term Forecasting of CO2 Emission Intensity in Power Grids by Machine Learning." *Applied Energy*, vol. 277, 2020, doi.org/10.1016/j.apenergy.2020.115351.

Maji, Diptyaroop, et al. "CarbonCast: Multi-Day Forecasting of Grid Carbon Intensity." *Proceedings of the 9th ACM International Conference on Systems for Energy-Efficient Buildings, Cities, and Transportation*, ACM, 2022, pp. 198-207, doi.org/10.1145/3563357.3564079.

Open-Meteo Historical Weather API. Open-Meteo, open-meteo.com/en/docs/historical-weather-api. Accessed May 2026.

Radovanovic, Ana, et al. "Carbon-Aware Computing for Datacenters." *arXiv*, 2021, arxiv.org/abs/2106.11750.

Tian, Hongda, et al. "Improving Day-Ahead Grid Carbon Intensity Forecasting by Joint Modeling of Local-Temporal and Cross-Variable Dependencies Across Different Frequencies." *Proceedings of the AAAI Conference on Artificial Intelligence*, 2026, arxiv.org/pdf/2601.06530.

Yan, Leyi, et al. "EnsembleCI: Ensemble Learning for Carbon Intensity Forecasting." *Proceedings of the 16th ACM International Conference on Future and Sustainable Energy Systems*, ACM, 2025, doi.org/10.1145/3679240.3734630.